

2016.0 RANGE ROVER (LG), 303-07

GLOW PLUG SYSTEM – TDV6 3.0L DIESEL

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the glow plug system and operation, refer to the relevant Description and Operation section of the workshop manual.

REFER to: [Glow Plug System](#) (303-07C Glow Plug System - TDV6 3.0L Diesel, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle.

NOTE:

Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

- 1. Verify the customer concern.
- 2. Visually inspect for obvious signs of electrical damage.

Visual Inspection

ELECTRICAL
▪ Glow plug warning indicator ▪ Fuses ▪ Glow plug relays ▪ Engine management control relay ▪ Wiring harness(es) ▪ Electrical connector(s) ▪ Glow plug(s) ▪ Engine Control Module (ECM)

- 3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
- 4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Poor starting (extreme weather conditions)	▪ Glow plugs inoperative/inefficient ▪ Fuel temperature too low	▪ Refer to the electrical circuit diagrams and test the glow plugs circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check for glow plug DTCs
High cold-engine emissions	▪ After-glow phase inoperative	

		NOTE: After-glow is designed to function at engine temperatures below 50°C (122°F), and below 2,500rpm - Refer to the electrical circuit diagrams and test the glow plugs circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check for glow plug DTCs
High cold-engine noise, vibration or harshness	After-glow phase inoperative	NOTE: After-glow is designed to function at engine temperatures below 50°C (122°F), and below 2,500rpm - Refer to the electrical circuit diagrams and test the glow plugs circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check for glow plug DTCs
Glow plug indicator illuminated permanently with no related DTCs set	If the end of line checks are not completed successfully and the glow plugs are not enabled, the glow plug indicator will be illuminated permanently when the vehicle is running	Using the manufacturer approved diagnostic system, perform routine - Powertrain - Enable/disable glow plugs - Located under 'Service Functions'. Retest

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this

vehicle, please refer to Section 100-00.

REFER to: [Diagnostic Trouble Code Index - TDV6 3.0L Diesel - Gen 2, DTC: B10A2-01 to P02D7-32](#) (100-00 General Information, Description and Operation).